Forecasting or Remembering?

A look-ahead test of language-model stock prediction from news headlines: same headlines, same target, four prompts that differ only in what the model is allowed to see.

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ABSTRACT Qwen2.5-7B and Qwen2.5-32B scored 18,144 FNSPID headlines on 124 large US stocks and ETFs (2012 to mid-2020) four ways: with the date, firm and headline; without the date; with the firm's name hidden; and with no headline. Ranking the sign of the same-session abnormal return, the full prompt reaches an AUC of 0.565 and 0.570, the name-hidden prompt 0.555 and 0.561, and the no-headline prompt 0.507 and 0.509. Hiding the name costs about 0.01 of AUC, which bounds what knowledge of the firm, or memory of the outcome, can add. Most of the signal sits in the 10% of headlines that report the move itself (AUC ≈ 0.72); on the next trading session the AUC is 0.505 to 0.508 and no interval excludes 0.5. On 20,262 time-stamped Benzinga headlines, every prompt that reads the headline loses about 0.02 of AUC from 2019–23 to 2025–26, the name-hidden prompt included. One model family cannot say whether that drop is memorised headline text or a changed market.

01

Summary

Papers report that language models reading headlines can sort stocks into winners and losers. The worry is that the model was trained on text written after the headline and simply remembers how the stock moved. This report separates the two by changing only what the model sees.

0.570	0.509	0.507	−0.021
Same-session AUC, full prompt, Qwen2.5-32B, FNSPID (7B: 0.565)	Same-session AUC with no headline at all, 32B (7B: 0.507)	Next-session AUC, full prompt, 32B; interval 0.498–0.516	Change in AUC after the training cutoff, 32B, headlines that do not report a move

KEY FINDINGS

1. **The headline carries the signal.** Removing it drops the same-session AUC from 0.565 to 0.507 (7B) and from 0.570 to 0.509 (32B). Without a headline the model is a coin flip within the noise.
2. **The firm's name adds about a hundredth.** Hiding it costs +0.011 [+0.006, +0.016] (7B) and +0.009 [+0.002, +0.016] (32B). The date adds +0.0004 (not distinguishable from zero) and +0.003. This bounds what knowledge of the firm, or memory of the outcome, can contribute.
3. **Most of the edge is reading the outcome.** The 10.1% of headlines that report a price move reach 0.712 and 0.724; everything else reaches 0.545 and 0.549, an upper bound because the filter misses about half the move reports.
4. **Nothing survives to the next session.** Next-session AUCs are 0.508 and 0.507, and no gap between prompts is distinguishable from zero on any subset.
5. **After the cutoff, everything that reads the headline drops by the same amount.** On Benzinga headlines that do not report a move, the full prompt loses 0.019 and 0.021 and the name-hidden prompt 0.022 and 0.017. The design cannot tell memorised text from a changed market.

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Scope and grade of evidence

Stream	Data	Grade	Supports
Four-prompt run	FNSPID, 18,144 headlines, 124 tickers, 2012-01 to 2020-06	Real data; one deterministic run per model (temperature 0); day-clustered bootstrap intervals	Findings 1–2
Move / next-session split	Same scores; hand-audited move filter	Real data; filter precision ≈ 93%, recall ≈ 50–55% on audit samples	Findings 3–4
Pre- vs post-cutoff	Benzinga via Alpaca, 9,223 + 11,039 headlines	Real data; the training cutoff is unpublished, so a 2023-07 to 2024-12 gap is excluded; coverage thin before 2021	Finding 5
Long-short Sharpe	Same scores	Illustration only: no intervals; the same-session version is not tradeable	Context for finding 4

Many subsets, prompts, targets and models were examined. A single cell with $p \approx 0.03$ is weak evidence and is labelled as such where it appears.

03

Data

FNSPID

Headlines come from the FNSPID news dataset (Hugging Face `sabareesh88/FNSPID_external`, 13.06 million rows). The sample keeps the most-covered tickers, headlines of 25 to 300 characters, and one randomly chosen headline per ticker per day, capped at 150 per ticker. Prices are split- and dividend-adjusted daily closes from Yahoo Finance. Twenty-three sampled tickers that have since been delisted (among them TWX, TIF, PXD and MYL) have no Yahoo history and drop out, leaving 124. Coverage for these tickers ends in June 2020.

About 4.4 million FNSPID rows carry a bare date stored as midnight UTC. Converting those to New York time would move them to 7 pm the previous day and put the event window a day before the news, so they are taken as calendar dates; rows with a real time of day are converted to New York time.

Benzinga, either side of the cutoff

For news the model cannot have read, timestamped Benzinga headlines were pulled from Alpaca's news API for the same tickers. 520 calendar days were sampled at random in each window, and every headline published that New York day was kept if it was tagged with three tickers or fewer, which drops market wraps. 215 of 60,162 items published outside their requested day were removed. Qwen2.5 was released in September 2024 and does not publish a training cutoff; estimates range from late 2023 to mid-2024, so the eighteen months from July 2023 to December 2024 are excluded.

Table 1. The three samples.

Sample	Dates	Labelled headlines	Tickers	Share up
FNSPID	2012-01 to 2020-06	18,144	124	0.504
Benzinga, pre-cutoff	2019-07 to 2023-06	9,223	119	0.489
Benzinga, post-cutoff	2025-01 to 2026-10	11,039	120	0.498

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Method

Table 2. Four prompts. Each ends "Answer with exactly one word: UP or DOWN." under the system prompt "You are an equity analyst. Respond with a single word."

Prompt	The model sees	A gap to the full prompt measures
`full`	date, firm, ticker, headline	(the number papers report)
`no_date`	firm, ticker, headline	how much the date helps recall
`masked`	headline with the firm's name and ticker replaced by "the company"	what knowing the firm adds
`no_news`	date, firm, ticker; no headline	pure memory; should be 0.5

The target is the sign of the abnormal return, the stock's return minus SPY's, from the close of the trading day before the news date to the close of the first trading day on or after it, with |abnormal return| below 50%:

$$y = 1[\ (P_{i,t_1}/P_{i,t_1-1} - 1) - (P_{SPY,t_1}/P_{SPY,t_1-1} - 1) > 0\ ] \tag{1}$$

Two variants reuse the same scores. The timestamp-correct target (Benzinga only) takes the first session whose 16:00 ET close is after the timestamp, so news released after the close moves to the next session. The next-session target runs from the close of t_1 to the close of t_1+1.

Each prompt is scored with vLLM 0.33 at temperature 0 and a single output token. The score is read from the top-20 log-probabilities:

$$p = P(UP) \,/\, (P(UP) + P(DOWN)) \tag{2}$$

No prompt in any of the 16 model × prompt × dataset runs lacked an UP or DOWN token. The metric is ROC AUC, the probability that a headline followed by an up move gets a higher p than one followed by a down move. Intervals are 95% percentile intervals from a bootstrap that resamples whole trading days (1,000 draws), so headlines on the same day keep their shared market shock. All prompts and both models are resampled on the same draws, which gives every gap its own interval; the two Benzinga windows are resampled independently. Long-short Sharpe ratios rank each day's headlines by p, demeaned within day, on days with at least three headlines.

05

What carries the signal

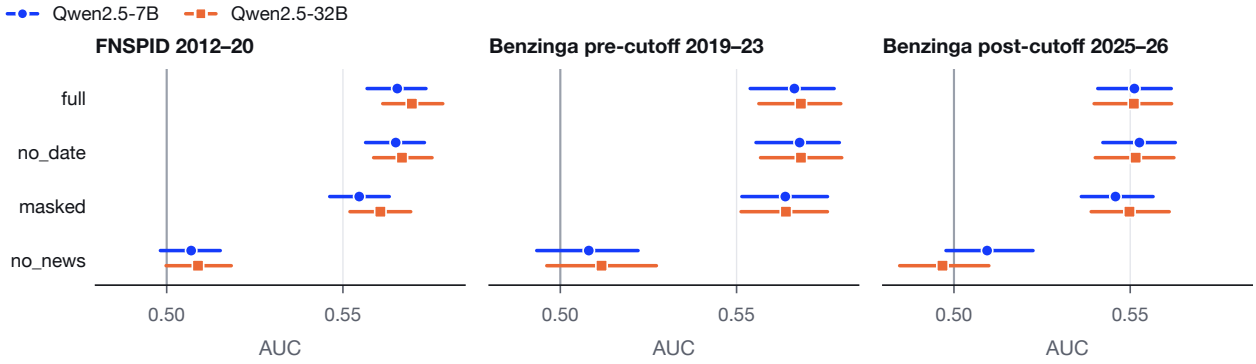


Figure 1. AUC for the sign of the same-session abnormal return, by prompt, model and sample. Dots are point estimates; lines are 95% day-clustered bootstrap intervals; the grey line is 0.5.

Source: results/phase_a/auc_table.csv; results_alpaca/phase_b/auc.csv

Table 3. Same-session AUC [95% interval], all headlines.

Sample	Model	full	no_date	masked	no_news
FNSPID	7B	0.565 [0.557, 0.574]	0.565 [0.556, 0.573]	0.555 [0.546, 0.563]	0.507 [0.498, 0.515]
FNSPID	32B	0.570 [0.561, 0.578]	0.567 [0.559, 0.575]	0.561 [0.552, 0.569]	0.509 [0.500, 0.518]
Benzinga pre	7B	0.566 [0.554, 0.578]	0.568 [0.556, 0.579]	0.564 [0.551, 0.576]	0.508 [0.493, 0.522]
Benzinga pre	32B	0.568 [0.556, 0.580]	0.568 [0.557, 0.580]	0.564 [0.551, 0.576]	0.512 [0.496, 0.527]
Benzinga post	7B	0.551 [0.541, 0.562]	0.553 [0.542, 0.563]	0.546 [0.536, 0.556]	0.509 [0.498, 0.522]
Benzinga post	32B	0.551 [0.540, 0.562]	0.552 [0.540, 0.562]	0.550 [0.539, 0.561]	0.497 [0.485, 0.510]

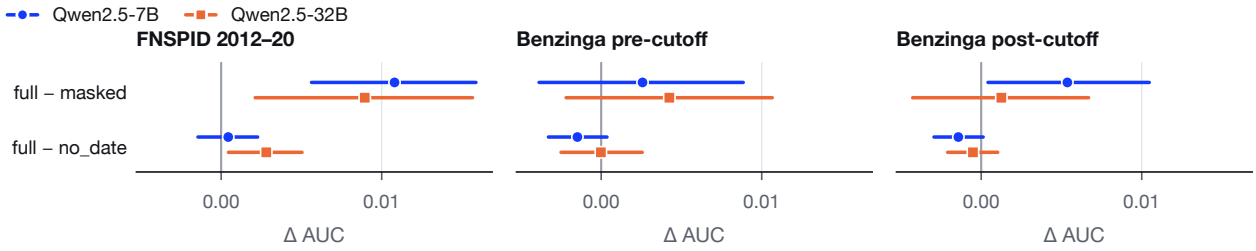


Figure 2. Paired AUC gaps between prompts, same-session target. Computed on the same bootstrap draws, so the interval is for the difference itself.

Source: results/phase_a/gaps.csv; results_alpaca/phase_b/gaps.csv

On FNSPID, hiding the name costs +0.0108 [+0.0056, +0.0159] for 7B and +0.0089 [+0.0021, +0.0157] for 32B; the date is worth +0.0004 [−0.0014, +0.0023] and +0.0028 [+0.0005, +0.0050]. On Benzinga the name gap is smaller and only the 7B post-cutoff interval excludes zero (+0.0054 [+0.0004, +0.0105]). The 32B model beats 7B on the masked prompt (+0.0060 [+0.0025, +0.0093]) but not detectably on the full prompt (+0.0041 [−0.0015, +0.0099]).

The name gap is an upper bound on memory, not a measurement of it: knowing the firm can help honestly, because the same news means more for some firms than for others. Glasserman and Lin (2023) ran the same masking test on GPT-3.5 and found that anonymised headlines did better in-sample, which they attributed to the firm's name distracting the model. A different model, a long-short return instead of an AUC, and a different horizon could each explain the opposite sign here.

06

Reading versus forecasting

The target window ends at the close of the session in which the news broke, so a headline written that afternoon may report the move it is asked to predict. A word filter, refined over four rounds of hand-checked random samples, flags 1,840 headlines (10.1%) that report a price move. On the final round about 93% of flagged headlines were genuine move reports, and roughly half of all move reports were caught.

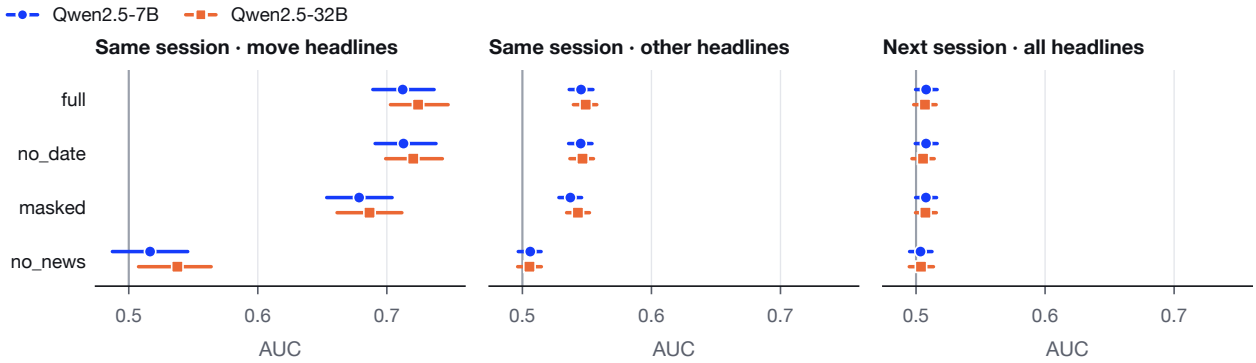


Figure 3. FNSPID AUC by headline type and target window. Move headlines n = 1,840; other headlines n = 16,304; next session n = 18,144.

Source: results/phase_a/auc_table.csv

On move headlines the full prompt reaches 0.712 [0.689, 0.737] and 0.724 [0.703, 0.747]; on the rest, 0.545 [0.536, 0.555] and 0.549 [0.539, 0.558]. On the next session, every prompt sits between 0.503 and 0.508 and every gap interval includes zero. With no headline, the 32B model reaches 0.538 [0.508, 0.564] on move days, which may mean it recalls which days were large for which firms; it is one of many cells examined and is reported as a lead, not a finding.

Table 4. Daily long-short Sharpe ratio, annualised (no intervals computed).

Model · prompt	Same session, all	Next session, all	Same session, other	Next session, other
7B · full	3.58	0.72	2.55	0.44
7B · masked	3.59	0.46	2.45	0.24
7B · no_news	0.97	0.34	0.66	0.69
32B · full	4.21	0.67	2.91	0.24
32B · masked	3.66	0.76	2.71	0.33
32B · no_news	1.30	0.85	0.90	0.75

The same-session portfolio is not tradeable: news released after the close is scored against a session that already ended. The no_news rows, which see no headline, show the noise level.

07

Before and after the cutoff

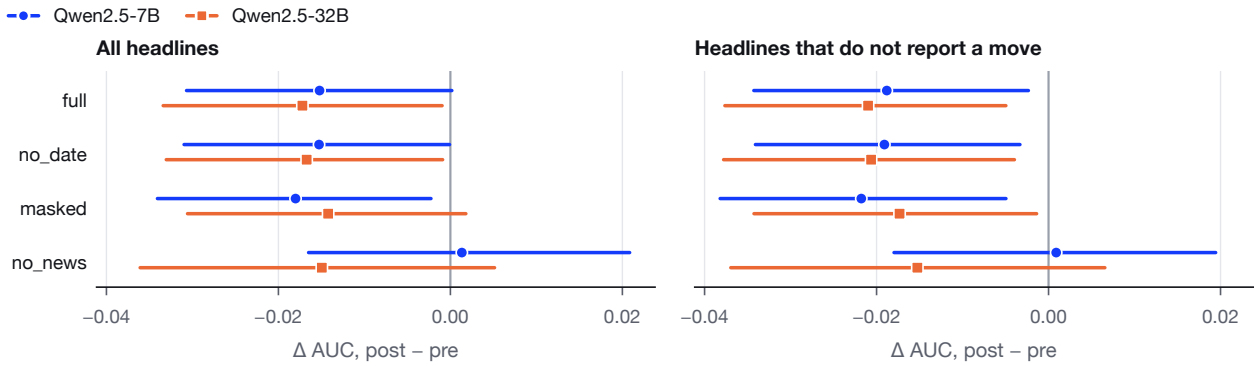


Figure 4. Benzinga: change in same-session AUC, 2025–26 minus 2019–23. Pre n = 9,223 (8,894 without move headlines); post n = 11,039 (10,556).

Source: results_alpaca/phase_b/gaps.csv

Table 5. Post − pre change in AUC [95% interval], same-session target.

Quantity	7B, all	32B, all	7B, no move	32B, no move
AUC full	−0.015 [−0.031, +0.000]	−0.017 [−0.033, −0.001]	−0.019 [−0.034, −0.002]	−0.021 [−0.038, −0.005]
AUC no_date	−0.015 [−0.031, −0.000]	−0.017 [−0.033, −0.001]	−0.019 [−0.034, −0.003]	−0.021 [−0.038, −0.004]
AUC masked	−0.018 [−0.034, −0.002]	−0.014 [−0.031, +0.002]	−0.022 [−0.038, −0.005]	−0.017 [−0.034, −0.001]
AUC no_news	+0.001 [−0.016, +0.021]	−0.015 [−0.036, +0.005]	+0.001 [−0.018, +0.019]	−0.015 [−0.037, +0.007]
Gap full − masked	+0.003 [−0.005, +0.011]	−0.003 [−0.012, +0.005]	+0.003 [−0.006, +0.011]	−0.004 [−0.012, +0.004]

The timestamp-correct target gives the same pattern (32B, no move: full −0.018 [−0.035, −0.002], masked −0.017 [−0.034, −0.002]). On the next session, every window AUC is between 0.491 and 0.513 and every change interval includes zero.

Two readings fit, and this design cannot separate them. If the model memorised headline text, hiding the name would not stop it recognising a 2021 sentence and the story that followed, so the masked prompt would fall with the full one, as it does. If headline style, story mix or the market changed between the windows, every prompt that reads the headline would fall by the same amount with no memory involved at all.

08

Robustness and limitations

Checks

- **Noise floor.** Random scores on the FNSPID labels give an AUC of 0.500 with a standard deviation of 0.0042 across 300 draws.
- **Calibration does not matter for AUC.** Mean P(UP) differs widely by prompt (7B: full 0.49, masked 0.70, no_news 0.01), which distorts accuracy but not ranking.

- **Masking quality.** Masking changed 33.2% of FNSPID headlines. An automatic check finds a residual identifying word in 12.1% of masked rows; in a 30-row sample about half were genuine leaks (for example "the company Depot") and half ETF themes such as "China". The full − masked gap is the same on leaky and clean rows (7B +0.0109 vs +0.0102; 32B +0.0085 vs +0.0083), so leakage does not drive it.

Limitations

- **Model vintage.** Qwen2.5-7B and 32B, released in 2024. An older model is what makes a two-year post-cutoff window possible, but the results describe these models, not the GPT-class models used in most papers or current releases.
- **Overlapping target.** The same-session window overlaps the news, and the move filter catches about half the reports, so "other headlines" overstates non-move signal.
- **After-close timing.** About two-thirds of the full FNSPID dataset carries a time of day; news after the close is scored against a session that already ended. The Benzinga timestamp check gives the same pattern.
- **Survivorship and universe.** Delisted tickers drop out; the universe is large, liquid, mostly still-listed names.
- **Different samples.** The Benzinga rule against multi-ticker items leaves 4% move headlines (FNSPID 10%) and 73% masked-changed (FNSPID 33%). Compare patterns across samples, not levels. Benzinga coverage is sparse before 2021, so the pre window leans toward 2021–23.

09

What would settle it

1. **A model that has read 2025–26.** Score the same windows with a model trained through the post-cutoff period, such as Qwen3.8-27B (released August 2026). If its 2025–26 AUC does not fall, the Qwen2.5 drop is memory; if it falls as well, the market or the headline mix changed.
2. **Time-stamped returns.** Minute bars would split each headline into before and after its own publication, replacing a filter that catches half of the move reports.
3. **A better move classifier.** A model-based "this headline reports the move" label, audited the same way, would tighten the bound on non-move signal.

Related work

Glasserman, P. and Lin, C. (2023). Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis. arXiv:2309.17322. Lopez-Lira, A. and Tang, Y. (2023). Can ChatGPT forecast stock price movements? Return predictability and large language models. arXiv:2304.07619; its results are on next-day returns with a different model.

A

Data and reproducibility

PROJECT	Local repository lookahead-probe: commits 82a043d (core runs, robustness), 1b7c117 (pre/post-cutoff test), 3763f72 (results pack). Not public: it contains licensed headline text and prices.
BRIEF	The four prompts, the target, the dedupe rule and the metrics were fixed in advance; deviations (the midnight-UTC date fix, a pandas 3 compatibility fix, delisted tickers) are logged in CHANGES.md.
DATA	FNSPID (Hugging Face sabareesh88/FNSPID_external); Benzinga headlines via the Alpaca news API, fetched 10 Nov 2026; daily adjusted prices from Yahoo Finance. Raw text and prices are not redistributed.
COMPUTE	Two Runpod pods on 10 Nov 2026: an A100 SXM 80GB (FNSPID, 01:02–02:01 UTC) and an A100 80GB PCIe (Benzinga, 02:24–03:07 UTC), about $3.10 in total. vLLM 0.33.0, torch 2.14.0+cu130, bf16 weights.
RESULTS FILES	final/results/scored_*.parquet and summary_*.csv; results/phase_a/{auc_table, gaps, sharpe}.csv; results_alpaca/phase_b/{auc, gaps}.csv; results/pack/RESULTS.md.
THIS REPORT	All tables are copied from the results CSVs (cross-checked by script); the four figures are redrawn from the same files by charts.py.

Ajai Upadhyaya · esnupi Publications No. 31 · November 2026 · Set in Charter, Archivo Black and Courier New. Code, runs, analysis and drafts were produced with Claude under the author's direction.